

Lwazi Magaba
CLDV POE Theory SUBMISSION

CLDV POE

Student Number:	St10434465
Programme Code:	BCA2
Module Lecturer:	Glynn Rudman
Module Code:	CLDV 6211
Date of Submission:	03-06-2026

I hereby declare that I did not plagiarise the content of this assignment and that this is my own work.

Assignment submitted via Turnitin: ☒ (Tick the Box)

Table of Contents

1	Cosmos DB vs Traditional SQL Databases	1
1.1	Azure SQL Databases	1
1.2	Azure Cosmos DB	2
2	Logic Applications and Sensitive Data	3
2.1	Event Grid and Workflow Automation	3
3	Serverless Computing vs Monolithic Applications	5
3.1	Serverless Architecture	5
3.2	Monolithic Applications	5
4	Reference List.....	7

1 COSMOS DB VS TRADITIONAL SQL DATABASES

Having both databases regarded as cloud-based infrastructures, they only have a single distinct difference, their designs as they fulfil different purposes (Brown, 2026). Of which are listed below:

Feature	Azure SQL Database	Azure Cosmos DB
Database Type	Relational Database making use of SQL	A NoSQL Database, making use of JSON
Data Structures	Data is collected and view into Tables and Rows	Data is stored in documents
Schemas	Fixed Schema, columns cannot be added or modified without affecting the entire table.	Schemas more flexible. Attributes can be added / removed per record saved into the document without affecting the entire database.
Relationships	Data is made relative to each other through the use of foreign keys that reference data from other tables.	Data is usually denormalized, data and attributes may not have a direct relationship with each other.
Scaling	Data can be modified horizontally (increasing resources) and vertically(Adding more servers). Allowing for data to be retrieved more dynamically(GeeksforGeeks, 2019).	Operates on global horizontal scaling whereby data is distributed across machines and devices globally(Brown, 2026).

1.1 Azure SQL Databases

Azure SQL Databases store information using tables and relationships, Examples include banking systems, booking systems, event management systems, etc. The EventEase web application is an example of that same system benefiting from

relational data because relationships are clearly defined between bookings, venues and events (Microsoft, 2022).

1.2 Azure Cosmos DB

Azure Cosmos DB is a NoSQL database service that is globally distributed, with its primary purpose being that it stores information as a JSON document (Brown, 2026). Here is an example below:

This database service is typically used for technologies that include social media platforms, IoT solutions, real-time analytics and global applications. The flexible schema allows different documents to contain different attributes / fields without requiring the entire database to be re-designed (Brown, 2026).

2 LOGIC APPLICATIONS AND SENSITIVE DATA

Logic applications are applications that allow organizations to integrate cloud services that don't require unnecessary coding practices and help automate workflows (Microsoft, 2024).

An example includes:

1. Customer submits a booking.
2. The logic app will receive the event.
3. After the event is received the confirmation email is generated.
4. The administrator is notified.

Although the application does provide benefits in automation, security considerations have to be emphasized significantly.

Security Considerations	
Authentication	Logic apps need to make use of directory authentication or managed identities whenever possible to avoid unauthorised users from exploiting the application (msmbaldwin, 2026).
Access Control	Implementing Role-Based Access Control to restrict access to unauthorised persons (msmbaldwin, 2026).
Encryption	All sensitive Information should be encrypted both in transmission using HTTPS and at rest within the Azure services (msmbaldwin, 2026).
Secret Management	Sensitive information (passwords, connection strings, etc) should be stored in Azure Key Vault instead of the application's codebase (msmbaldwin, 2026).
Monitoring	Azure Monitor and diagnostics should be used to track any workflow executions that occur and detect any suspicious activity (Microsoft, 2024).

2.1 Event Grid and Workflow Automation

Azure Event Grid a service that focuses on event-routing. It is a service that enables applications to react to any events as they occur in real-time. Instead of checking for updates repeatedly, they respond immediately when the events are generated (spelluru, 2024).

Here is an example with a system that handles Image Processing:

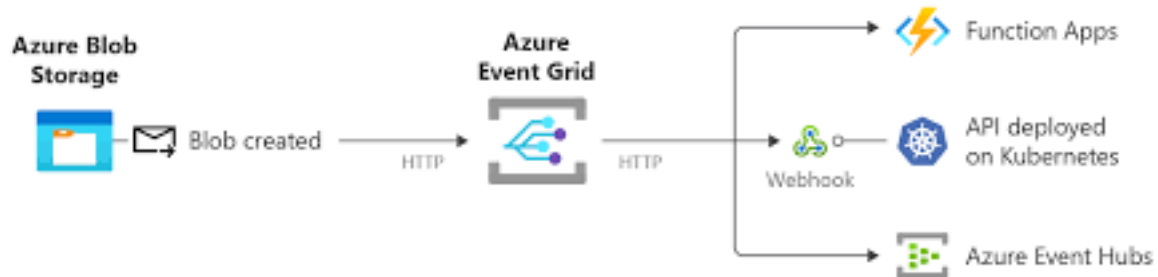


Figure 1 A workflow detailing receive events from Azure services triggered when images are uploaded to blob storage (robece, 2025)

Benefits of this Event Grid include

- Scalability- as large amounts of events are handled efficiently (robece, 2025).
- Efficiency- Constant polling (Tracking long-running operations) is eliminated (Imazuel, 2026).
- Decoupling- Services can operate independently of each other, mitigating any instances of dependency issues that make use of the entire infrastructure (kexugit, 2019).
- Reliability – Supports event-driven architectures (polish subscribe / Event streaming) (EdPrice-MSFT, 2026).

Combining Azure Functions, Logic applications, Azure Storage Accounts, databases and Event Grid, automated cloud workflows can be established. Providing organizations with appropriate production and deployment efficiency (Microsoft, 2024).

3 SERVERLESS COMPUTING VS MONOLITHIC APPLICATIONS

Cloud environments make use of serverless technologies on an increasing basis, making use of one of two architectures (GeeksforGeeks, 2024).

3.1 Serverless Architecture

An architecture that enables serverless computing that allows programs to execute only when an event is triggered (GeeksforGeeks, 2024).. Examples include Azure functions, Azure Logic Apps and Azure Event Grid. With advantages that include:

- Autonomous Scaling – where Resources are scaled automatically based on traffic (GeeksforGeeks, 2024)..
- Reduced Costs – As developers pay for execution only(GeeksforGeeks, 2024)..
- Reduced Maintenance – Manual server management is not required (GeeksforGeeks, 2024).
- Event Driven System – Only executes when necessary, reducing resources used (GeeksforGeeks, 2024).

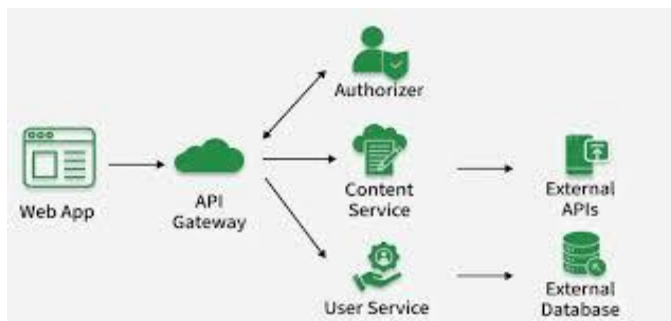


Figure 2 Image of a serverless architecture used in controlling cloud allocation and provision between cloud providers (GeeksforGeeks, 2024)

An example would include a booking confirmation system uses and Event Grid to detect a new booking, Azure Function to process said booking, and have logic apps to send confirmation emails without making use of a server that runs continuously (spelluru, 2024).

3.2 Monolithic Applications

A monolithic application typically contains a user interface, business logic and a data access layer, that all co-operate within one deployable application(Sayed, 2022).

The EventEase MVC application primarily follows a monolithic architecture as it makes use of all major components that are deployed together (Sayed, 2022).

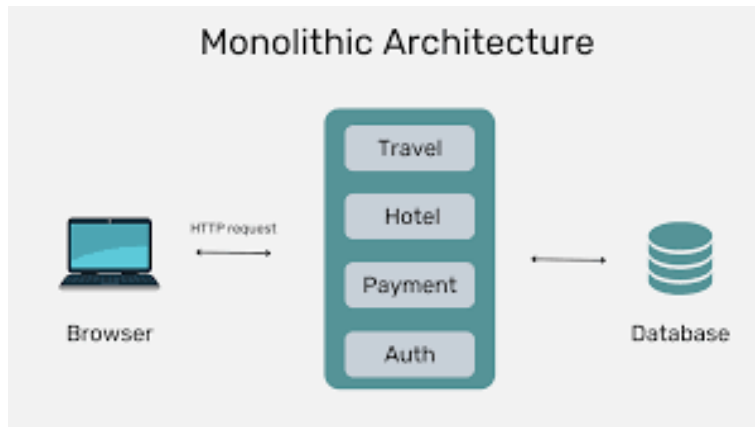


Figure 3 An image of a monolithic architecture that consists of different components such as database, server side, and client-side applications in one place (Sayed, 2022).

Cloud development environments provide various approaches to building applications that are scalable. Azure SQL Database and Cosmos DB address the different types of storage requirements that may be encountered within an organizations data handling needs, while Logic Apps and Event Grid help enable automated workflows that work effectively. And when combined with serverless computing services i.e. Azure Functions, organizations can establish highly scalable, cost-effective and secure cloud solutions to deploy into production.

4 REFERENCE LIST

- Brown, Mark.J. .2026. *Unified AI Database - Azure Cosmos DB*. [online] Microsoft.com. Available at: <https://learn.microsoft.com/en-us/azure/cosmos-db/overview> [Accessed 2 Jun. 2026].
- EdPrice-MSFT .2026. *Event-driven architecture style - Azure Architecture Center*. [online] learn.microsoft.com. Available at: <https://learn.microsoft.com/en-us/azure/architecture/guide/architecture-styles/event-driven> [Accessed 3 Jun. 2026].
- GeeksforGeeks .2019. *Horizontal and Vertical Scaling In Databases*. [online] GeeksforGeeks. Available at: <https://www.geeksforgeeks.org/dbms/horizontal-and-vertical-scaling-in-databases/> [Accessed 2 Jun. 2026].
- GeeksforGeeks. 2024. *Serverless Architecture*. [online] GeeksforGeeks. Available at: <https://www.geeksforgeeks.org/system-design/serverless-architectures/> [Accessed 2 Jun. 2026].
- kexugit .2019. *Azure - Event-Driven Architecture in the Cloud with Azure Event Grid*. [online] Microsoft.com. Available at: <https://learn.microsoft.com/en-us/archive/msdn-magazine/2018/february/azure-event-driven-architecture-in-the-cloud-with-azure-event-grid> [Accessed 2 Jun. 2026].
- Imazuel .2026. *azure.core.polling.base_polling module*. [online] Microsoft.com. Available at: https://learn.microsoft.com/en-us/python/api/azure-core/azure.core.polling.base_polling?view=azure-python [Accessed 2 Jun. 2026].
- Microsoft .2022. *What is the Azure SQL Database service? - Azure SQL Database*. [online] learn.microsoft.com. Available at: <https://learn.microsoft.com/en-us/azure/azure-sql/database/sql-database-paas-overview?view=azuresql> [Accessed 2 Jun. 2026].
- Microsoft .2024. *Overview - Azure Logic Apps*. [online] learn.microsoft.com. Available at: <https://learn.microsoft.com/en-us/azure/logic-apps/logic-apps-overview> [Accessed 3 Jun. 2026].

msmbaldwin .2026. *Microsoft cloud security benchmark - Identity Management*.

[online] learn.microsoft.com. Available at: <https://learn.microsoft.com/en-us/security/benchmark/azure/mcsb-identity-management> .

robece .2025. *Use cases for using Azure Event Grid - Azure Event Grid*. [online]

Microsoft.com. Available at: <https://learn.microsoft.com/en-us/azure/event-grid/use-cases> [Accessed 2 Jun. 2026].

Sayed, H. 2022. *Event-Driven Architecture Simplified: Monolith to Microservices*.

[online] Kubesimplify Blog. Available at: <https://blog.kubesimplify.com/event-driven-architecture-simplified-monolith-to-microservices> [Accessed 2 Jun. 2026].

spelluru 2024. *What is Azure Event Grid? - Azure Event Grid*. [online]

learn.microsoft.com. Available at: <https://learn.microsoft.com/en-us/azure/event-grid/overview> [Accessed 3 Jun. 2026].